Network Service Enumeration & Traffic Analysis Lab

Objective

Simulate a controlled network environment to identify services, analyze network traffic, and understand how systems communicate using TCP and HTTP protocols.

Tools Used

Nmap, Wireshark, Netcat, curl, Python HTTP Server

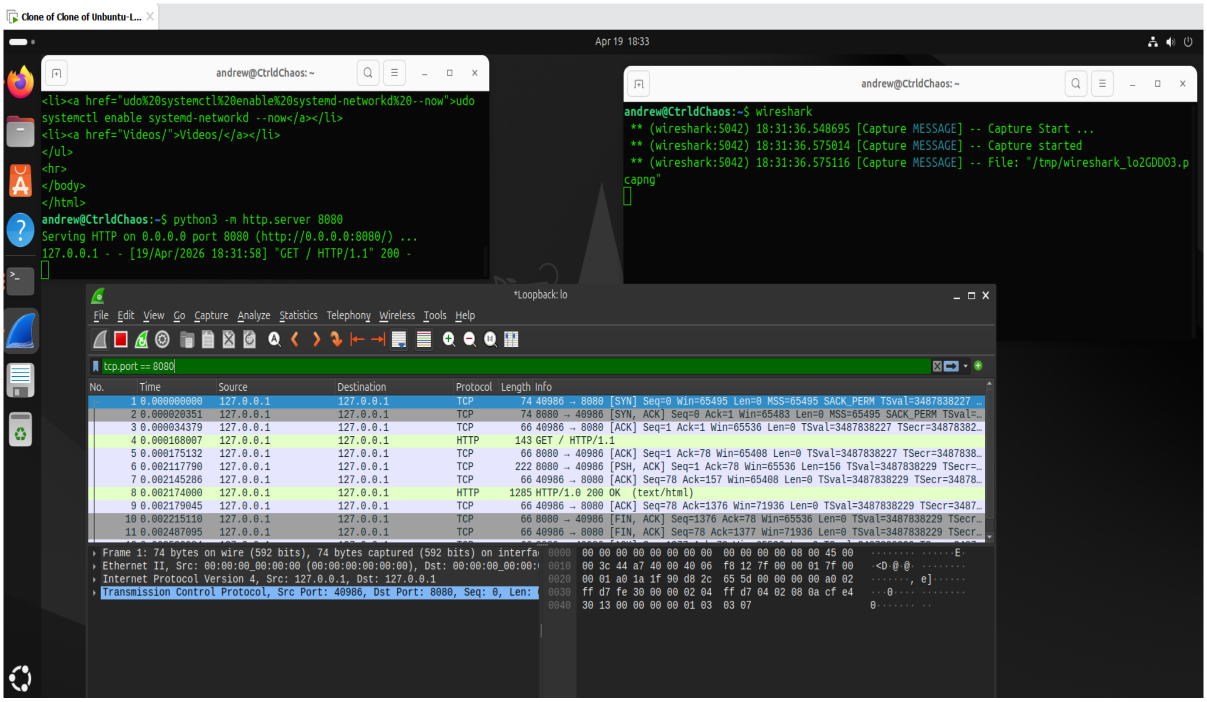
Methodology

1. Created a local HTTP server using Python. 2. Scanned system services using Nmap. 3. Observed server logs during active scanning. 4. Captured traffic using Wireshark. 5. Analyzed TCP handshakes and HTTP communication. 6. Simulated raw TCP communication using Netcat.

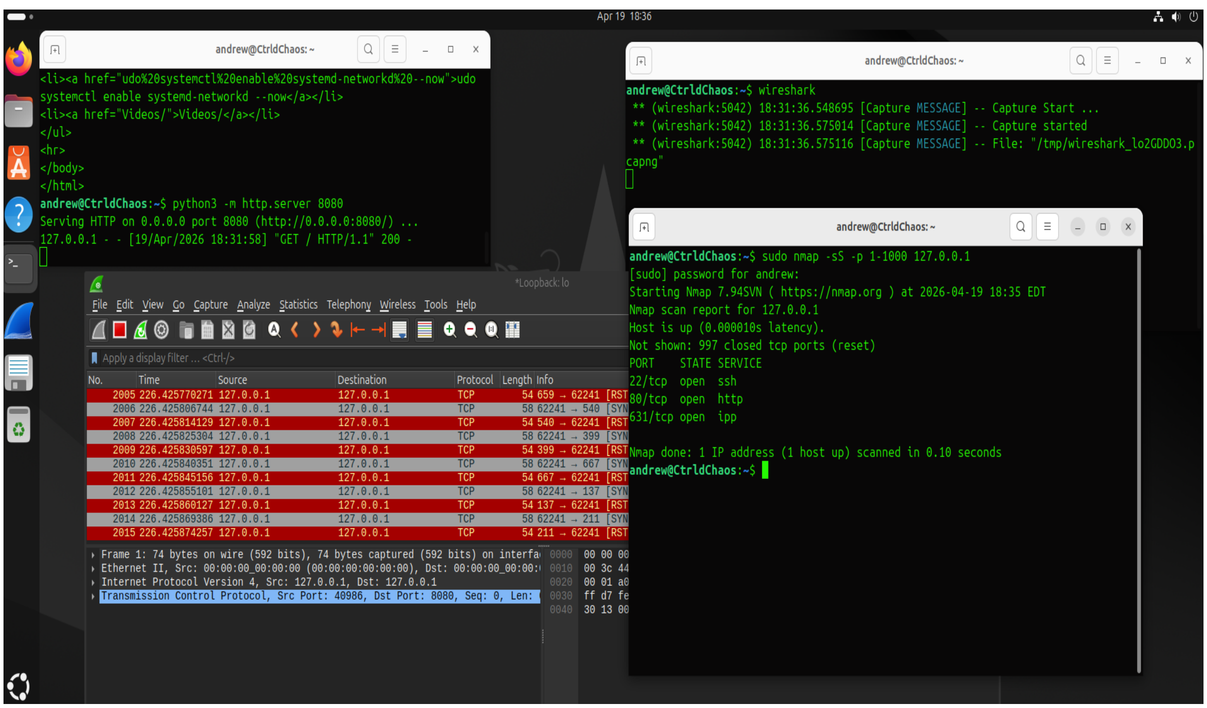
Key Findings

- Open ports and services were successfully identified. - Nmap actively probes web endpoints when scanning. - TCP follows a structured handshake process. - HTTP communication is readable and structured compared to raw TCP.

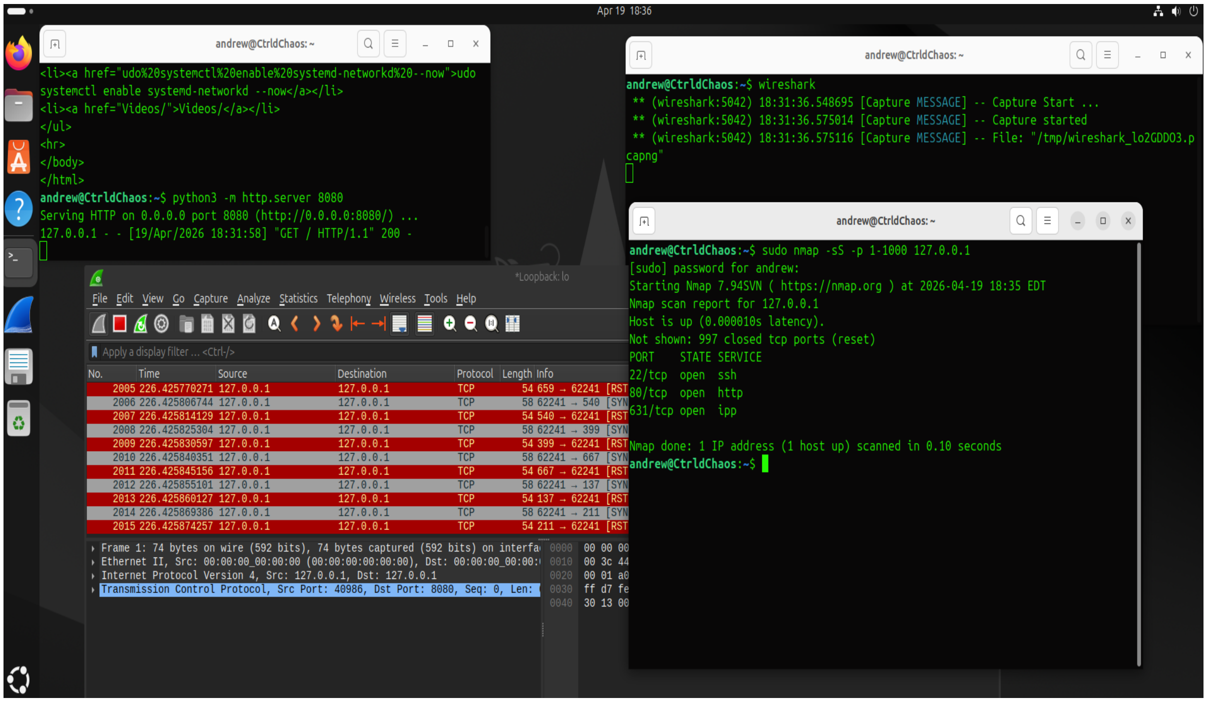
Evidence & Analysis



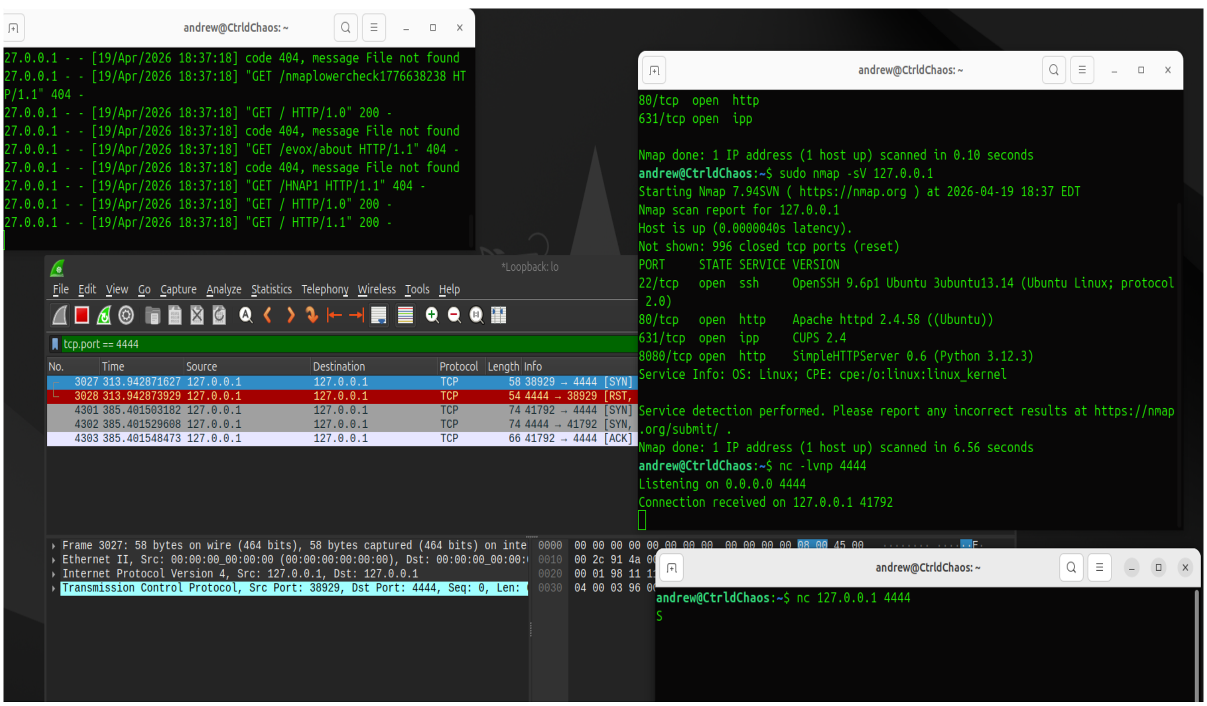
Wireshark capture showing HTTP request/response and TCP handshake.



Nmap scan activity observed in real-time within Wireshark.



Initial Nmap scan results identifying open ports.



Netcat connection demonstrating raw TCP communication.

Conclusion

This lab demonstrated the ability to identify services, analyze network behavior, and understand protocol communication. The combination of scanning, interaction, and packet analysis provides a strong foundation for cybersecurity and network analysis roles.

Resume Summary

Built and analyzed a virtual network lab to simulate service enumeration and traffic inspection using Nmap and Wireshark, capturing and analyzing TCP and HTTP communication flows.